

High ACAT1 expression in estrogen receptor negative basal-like breast cancer cells is associated with LDL-induced proliferation.

The specific role of dietary fat in breast cancer progression is unclear, although a low-fat diet was associated with decreased recurrence of estrogen receptor alpha negative (ER(-)) breast cancer. ER(-) basal-like MDA-MB-231 and MDA-MB-436 breast cancer cell lines contained a greater number of cytoplasmic lipid droplets compared to luminal ER(+) MCF-7 cells. Therefore, we studied lipid storage functions in these cells. Both triacylglycerol and cholesteryl ester (CE) concentrations were higher in the ER(-) cells, but the ability to synthesize CE distinguished the two types of breast cancer cells. Higher baseline, oleic acid- and LDL-stimulated CE concentrations were found in ER(-) compared to ER(+) cells. The differences corresponded to greater mRNA and protein levels of acyl-CoA:cholesterol acyltransferase 1 (ACAT1), higher ACAT activity, higher caveolin-1 protein levels, greater LDL uptake, and lower de novo cholesterol synthesis in ER(-) cells. Human LDL stimulated proliferation of ER(-) MDA-MB-231 cells, but had little effect on proliferation of ER(+) MCF-7 cells. The functional significance of these findings was demonstrated by the observation that the ACAT inhibitor CP-113,818 reduced proliferation of breast cancer cells, and specifically reduced LDL-induced proliferation of ER(-) cells. Taken together, our studies show that a greater ability to take up, store and utilize exogenous cholesterol confers a proliferative advantage to basal-like ER(-) breast cancer cells. Differences in lipid uptake and storage capability may at least partially explain the differential effect of a low-fat diet on human breast cancer recurrence.